SC Honors Geometry - Nappier

5.1 – 5.3 Proof Practice — Proving Triangles Congruent

Nine proofs, three parts, and the scaffolding comes off as you go. Every reason you need is in the bank below. Line 1 is always the given, word for word. The last line is always exactly what you were asked to prove.

REASON BANK — every reason on this page is one of these

Given · Reflexive Property of Congruence · Definition of midpoint · Definition of a segment bisector · Definition of angle bisector

Vertical Angles Theorem · Alternate Interior Angles Theorem · Definition of perpendicular lines · Right Angles Congruence Theorem · Definition of a right triangle

SSS Congruence Theorem · SAS Congruence Theorem · ASA Congruence Theorem · AAS Congruence Theorem · HL Congruence Theorem

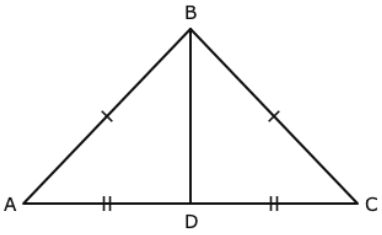
Part A — The statements are done. Supply each reason.

1. Two sides and the piece they share (5.3)

Given: $A\bar{B} \cong C\bar{B}$, and D is the midpoint of $A\bar{C}$

Prove: $\triangle ABD \cong \triangle CBD$

Three pairs of sides and no angles at all — that names itself.



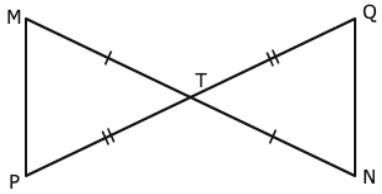
Statements	Reasons
1. $A\bar{B} \cong C\bar{B}$	**1.**
2. D is the midpoint of $A\bar{C}$	**2.**
3. $A\bar{D} \cong C\bar{D}$	**3.**
4. $B\bar{D} \cong B\bar{D}$	**4.**
5. $\triangle ABD \cong \triangle CBD$	**5.**

2. Segments that cut each other in half (5.3)

Given: $M\bar{N}$ and $P\bar{Q}$ bisect each other at T

Prove: $\triangle MTP \cong \triangle NTQ$

The angle you need is not given — it is sitting at the crossing.



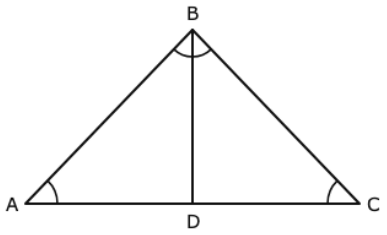
Statements	Reasons
1. $M\bar{N}$ and $P\bar{Q}$ bisect each other at T	**1.**
2. $M\bar{T} \cong N\bar{T}$ and $P\bar{T} \cong Q\bar{T}$	**2.**
3. $\angle MTP \cong \angle NTQ$	**3.**

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Statements	Reasons
4. △MTP ≅ △NTQ	**4.**

3. An angle bisector and a pair of angles (5.3)

Given: $\overline{BD}$ bisects ∠ABC, and ∠A ≅ ∠C
Prove: △ABD ≅ △CBD
Two angles and a side — check WHERE the side sits before you name it.

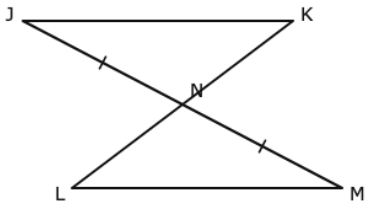


Statements	Reasons
1. $\overline{BD}$ bisects ∠ABC	**1.**
2. ∠ABD ≅ ∠CBD	**2.**
3. ∠A ≅ ∠C	**3.**
4. $\overline{BD} \cong \overline{BD}$	**4.**
5. △ABD ≅ △CBD	**5.**

Part B — The reasons are done. Write each statement.

4. Parallel sides and a midpoint (5.3)

Given: $\overline{JK} \parallel \overline{ML}$, and N is the midpoint of $\overline{JM}$
Prove: △JKN ≅ △MLN
Parallel lines hand you an angle pair for free.



Statements	Reasons
1.	**1.** Given
2.	**2.** Alternate Interior Angles Theorem
3.	**3.** Given
4.	**4.** Definition of midpoint
5.	**5.** Vertical Angles Theorem
6.	**6.** ASA Congruence Theorem

5. A right triangle pair (5.3)

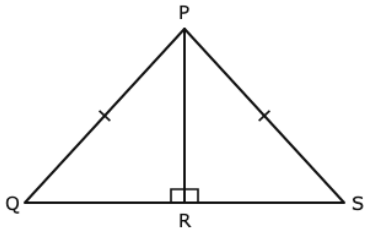
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Given: $P\bar{R} \perp Q\bar{S}$, and $P\bar{Q} \cong P\bar{S}$

Prove: $\triangle PQR \cong \triangle PSR$

One theorem on this page only works on right triangles. Earn the right angle first.



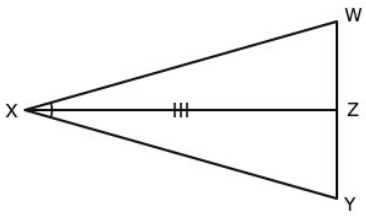
Statements	Reasons
1.	**1.** Given
2.	**2.** Definition of perpendicular lines
3.	**3.** Definition of a right triangle
4.	**4.** Given
5.	**5.** Reflexive Property of Congruence
6.	**6.** HL Congruence Theorem

6. Bisected angle, matching angles (5.3)

Given: $X\bar{Z}$ bisects $\angle WXY$, and $\angle W \cong \angle Y$

Prove: $\triangle WXZ \cong \triangle YXZ$

The shared side is not between the two angles. That decides the name.



Statements	Reasons
1.	**1.** Given
2.	**2.** Definition of angle bisector
3.	**3.** Given
4.	**4.** Reflexive Property of Congruence
5.	**5.** AAS Congruence Theorem

Part C — On your own. Statements AND reasons.

7. Two pairs of sides across a point (5.3)

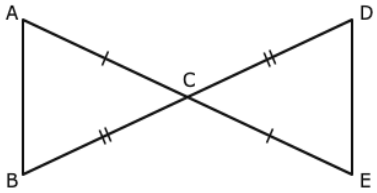
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Given: $A\bar{C} \cong E\bar{C}$, and $B\bar{C} \cong D\bar{C}$

Prove: $\triangle ABC \cong \triangle EDC$

What sits between the two given sides?



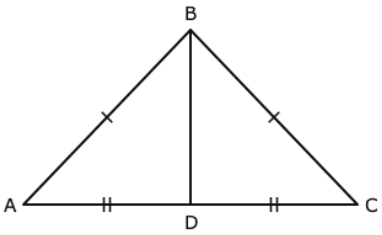
Statements	Reasons
1.	**1.**
2.	**2.**
3.	**3.**
4.	**4.**

8. The same figure, a different route (5.3)

Given: D is the midpoint of $A\bar{C}$, and $B\bar{D} \perp A\bar{C}$

Prove: $\triangle ABD \cong \triangle CBD$

A perpendicular gives you right angles; right angles are congruent to each other.



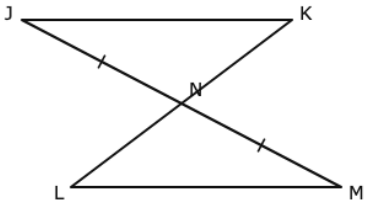
Statements	Reasons
1.	**1.**
2.	**2.**
3.	**3.**
4.	**4.**
5.	**5.**
6.	**6.**
7.	**7.**

9. Parallel and congruent (5.3)

Given: $J\bar{K} \parallel M\bar{L}$, and $J\bar{K} \cong M\bar{L}$

Prove: $\triangle JKN \cong \triangle MLN$

Both of the angle pairs you need come from the parallel lines and the crossing.



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Statements	Reasons
1.	**1.**
2.	**2.**
3.	**3.**
4.	**4.**
5.	**5.**